Any-ttach: Quick End-effector Swapping Enables Manipulation Dexterity with Simplicity

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DexLab — Duke Robot Dexterity Lab

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videos + details

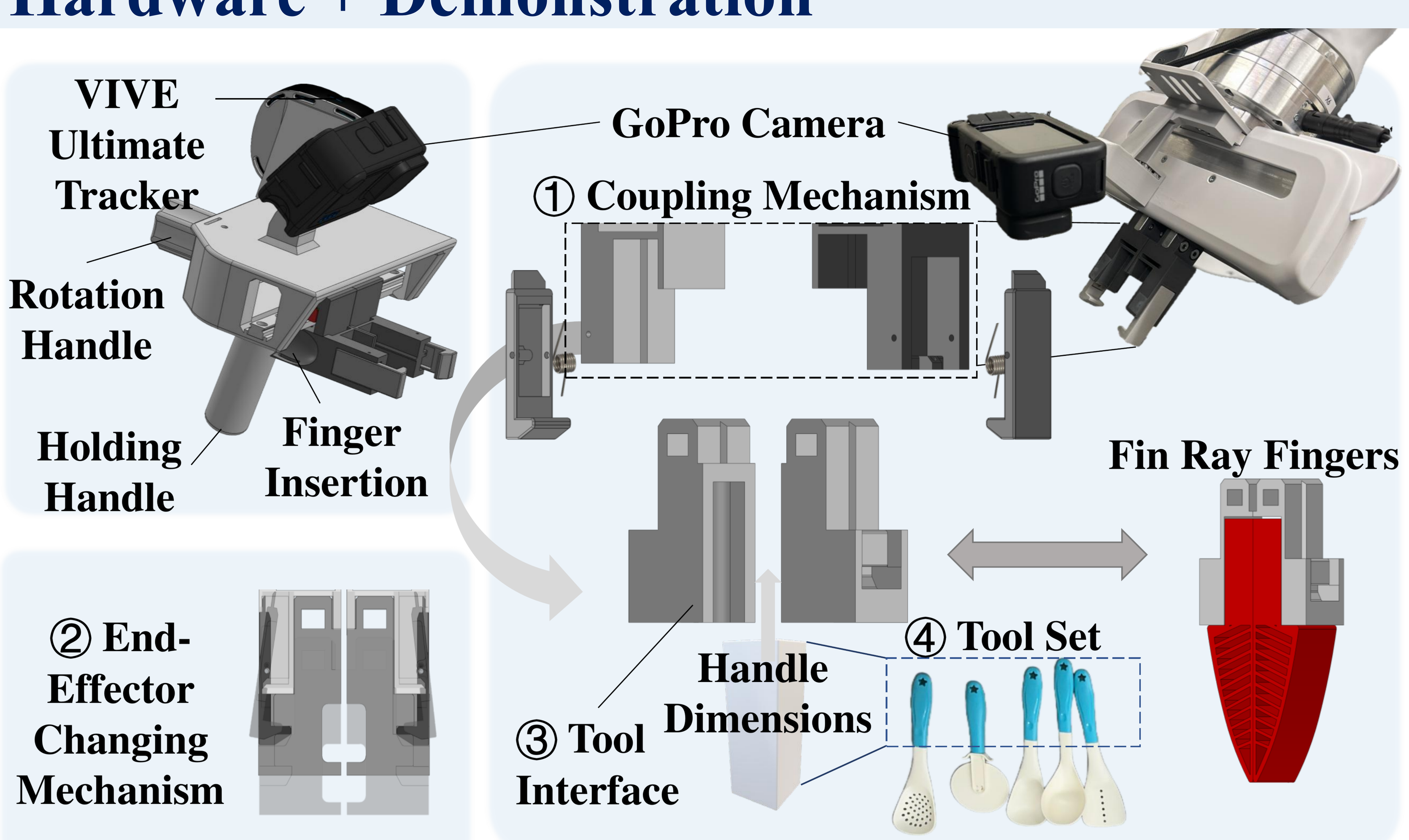
Introduction

Robotic dexterity is often pursued by adding mechanical complexity: high-DoF hands, dense sensing, and contact-rich control. **Any-ttach** takes a tool-centric alternative: task-specific contact geometry lives in interchangeable tools, not only in the hand. Our main contributions are:

(1) Automatic passive quick-swap interface for a simple parallel gripper.
(2) Shared robot and handheld coupling with demonstrations preserving tool geometry.
(3) Hierarchical tool-skill execution with planning, learned/planned skills, monitoring, and retry.
(4) Evaluation on 15 tools, 6 tool-use subskills, and 2 multi-tool long-horizon tasks.

Simple & Interchangeable | **Human Demonstration** | **Tool-Centric Dexterity**

Hardware + Demonstration



Shared coupling geometry links robot execution and handheld data collection.

Method

System

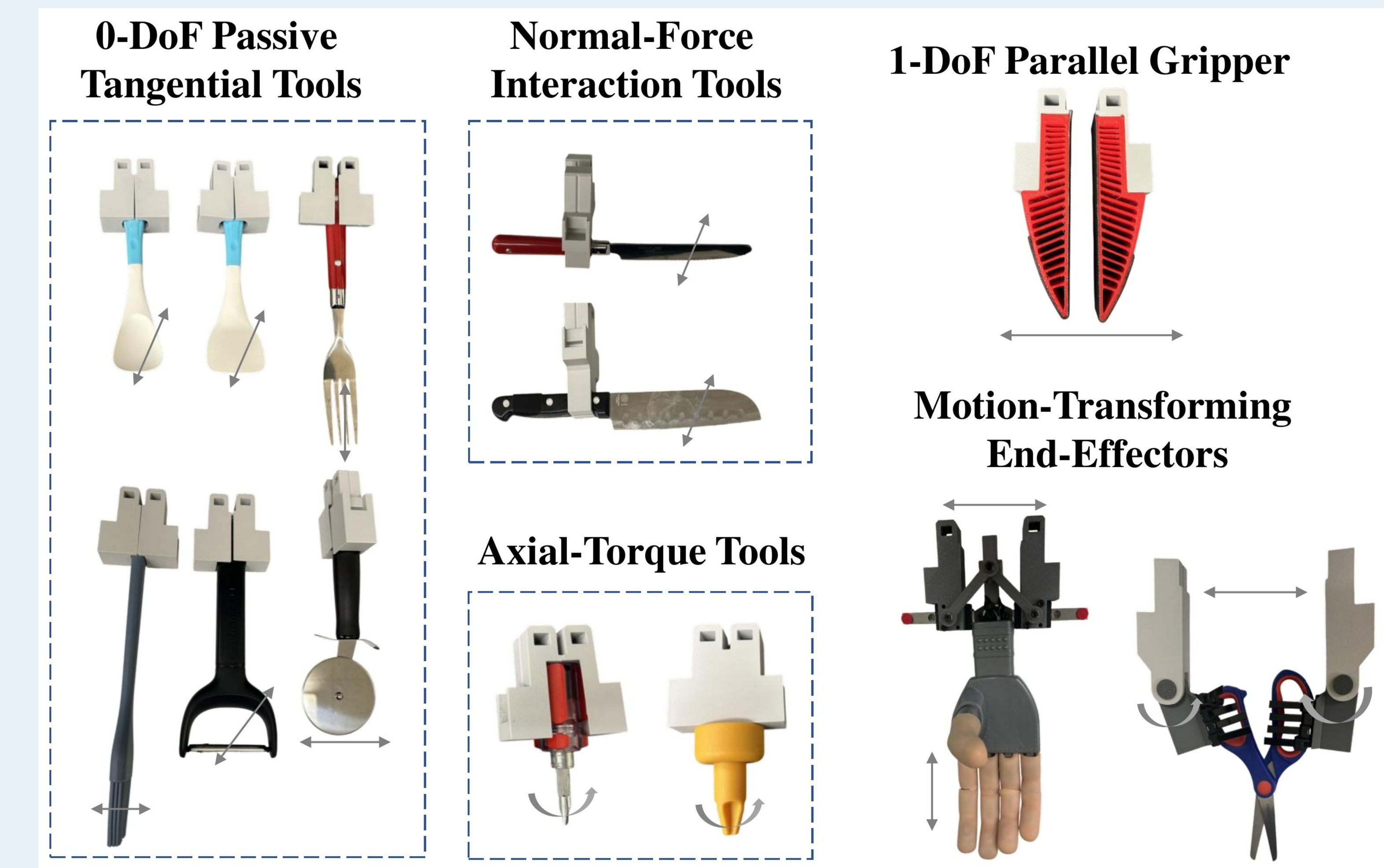
Task planner + skill execution + swap primitives + human demonstration.

Execution Loop

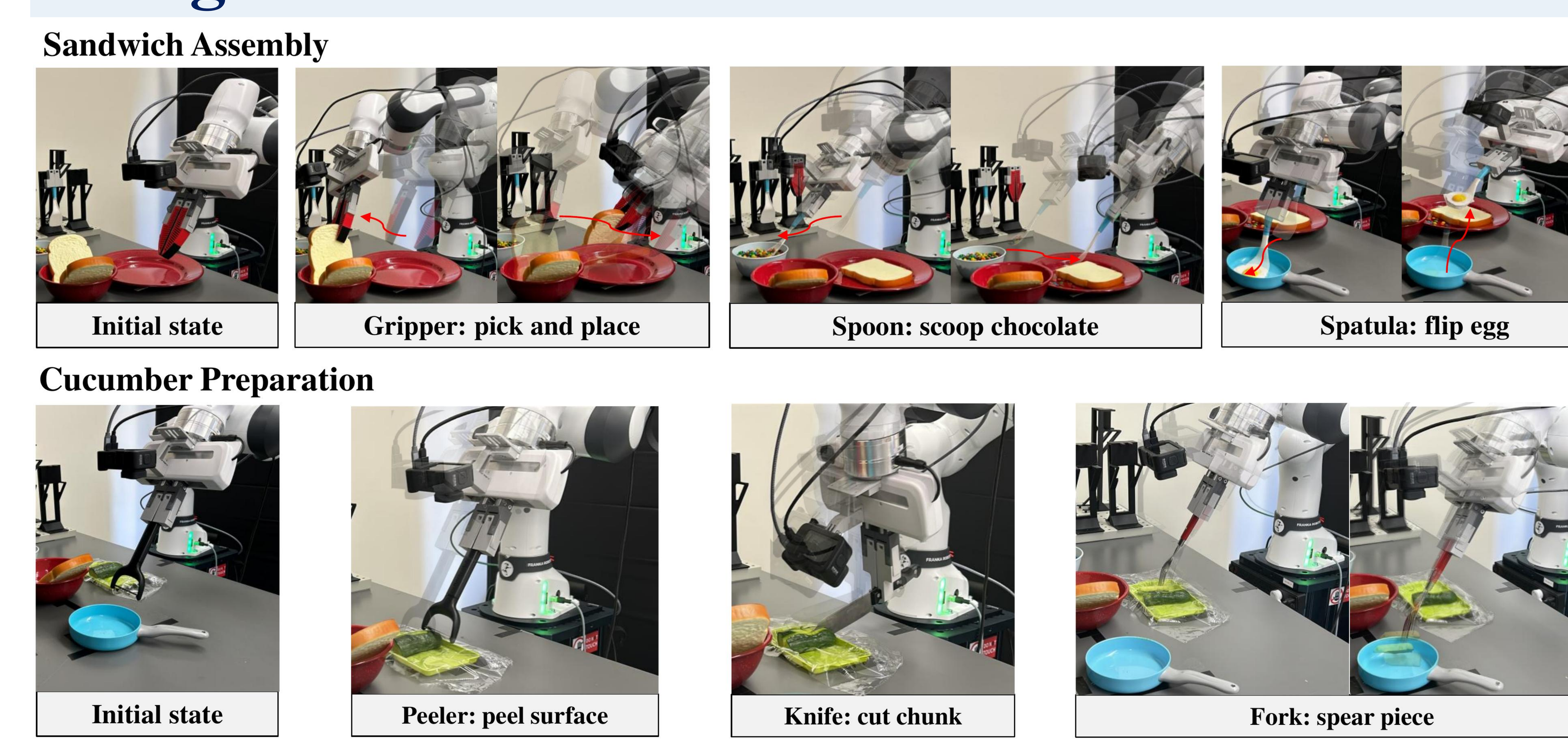
- VLM decomposes an instruction into an ordered tool-skill sequence.
- Robot attaches the planned module through a mechanically constrained dock.
- Skills run as learned Diffusion Policy controllers, pose-driven planners, or primitives.
- Vision-language monitoring checks preconditions and outcomes; failed steps can be retried.

Preliminary Results

Tool Coverage



Long-horizon Tasks



Sandwich Assembly and Cucumber Preparation compose autonomous tool changes with tool-conditioned skills.

Conclusion and Future Work

Any-ttach demonstrates a tool-centric path to robotic dexterity through fast, reliable end-effector swapping. Its mechanically constrained interface reduces grasp-induced variability and improves performance across demonstration collection, individual skills, and long-horizon tasks. By shifting task-specific dexterity from the robot hand into interchangeable tools, the system expands manipulation capability without requiring increasingly complex end-effectors. Future work will improve robustness under docking uncertainty, load/wear characterization, automatic adapter generation, and broader articulated-tool evaluation.

Dexterity by swapping end-effectors

The robot does not need to grasp every tool in a stable functional pose. It docks, locks, uses, detaches, and switches.